Review notes · for you

A connected world: unit check

IoT · Level 4 · Unit check

Materials
Paper and a pen. No device needed.

Learning goals

- How things talk
- Pass it on: mesh networks
- Keep connected things safe
- Your connected project

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Quiz sheet

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Name _____ Class _____ Date _____

1 **Twenty soil sensors on a farm 4 km from school run on batteries, and each sends a few bytes an hour. The school can't pay for monthly data plans. Which network suits them best?**

◯ Wi-Fi ◯ A mobile network ◯ LoRa

2 **A sensor that only uses 2G can stop working when its country switches off 2G, even though nothing on the sensor has broken.**

◯ True ◯ False

3 **Match each word to its meaning.**

Hop	• • • •	A radio that only passes messages on
Relay	• • • •	Doing the harmless thing when a part breaks
Gateway	• • • •	One step of a message from one device to the next
Botnet	• • • •	Passes a network's messages to the internet
Fail safe	• • • •	Hijacked devices controlled together

4 **Find the weak point in a line of sensors.**

Sensors stand 200 m apart along a river, and each radio reaches 250 m. A message from the sensor 1,000 m from the gateway needs _____ hops. If the sensor nearest the gateway breaks, every sensor beyond it is _____.

Word bank: 4 · rerouted · 5 · cut off

5 **Which would have kept Mirai out of a camera?**

☐ Switching off remote access it didn't need ☐ Covering the lens with tape ☐ Cleaning the lens
☐ Buying a camera with a sharper picture ☐ Changing its default password

6 **Put these in order of how far they reach, shortest first.**

☐ LoRa in a town
☐ LoRa in open country
☐ Bluetooth to a phone
☐ Wi-Fi indoors

7 **A school pump fills a water tank. Its level sensor stops answering. What should the pump do?**

◯ Pump at half speed ◯ Stop, and send an alert ◯ Keep pumping until someone checks

8 **When a device has been taken over by a botnet, its owner usually sees an obvious warning.**

◯ True ◯ False

9 **A team wants to know how busy the school gate is each hour. Which choices collect only what the job needs?**

☐ Sharing hourly totals, and deleting the raw counts after a week ☐ A camera that records faces all day
☐ A beam counter that counts people passing ☐ Recording the ID of each phone that passes

10 **Two designs score 88% and 86%. What is the best next step?**

◯ Pick the cheaper one, since the scores are so close ◯ Ask the people affected, and test other weights
◯ Build the 88% design: the scores have decided

Check your answers

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1 **LoRa**

4 km rules out Wi-Fi. A mobile network would reach, but twenty SIMs and data plans cost money every month and use more battery.

2 **True**

Many countries are switching off 2G and 3G to reuse the radio space for newer networks.

3 **Hop — One step of a message from one device to the next**
Relay — A radio that only passes messages on
Gateway — Passes a network's messages to the internet
Botnet — Hijacked devices controlled together
Fail safe — Doing the harmless thing when a part breaks

These five words carry the course: messages hop through relays to a gateway, botnets are where unchanged passwords lead, and failing safe protects people when parts break.

4 **Sensors stand 200 m apart along a river, and each radio reaches 250 m. A message from the sensor 1,000 m from the gateway needs 5 hops. If the sensor nearest the gateway breaks, every sensor beyond it is cut off.**

With 200 m spacing and 250 m of range, each sensor reaches only its neighbours, so 1,000 ÷ 200 = 5 hops.

5 **Changing its default password; Switching off remote access it didn't need**

Mirai logged in over the internet with factory passwords.

6 **Top to bottom: 3, 4, 1, 2**

About 10–30 m, 30–50 m, 2–5 km, and 10 km or more.

7 **Stop, and send an alert**

Without the sensor, the pump can't tell whether the tank is full, so pumping blind can overflow it, or burn out the pump if the well runs dry.

8 **False**

Botnets stay hidden so they can keep using the device: the camera still shows pictures and the router still works.

9 **A beam counter that counts people passing; Sharing hourly totals, and deleting the raw counts after a week**

The job needs one number per hour. Counting without pictures and keeping only totals answers it; faces and phone IDs identify people, and could leak.

10 **Ask the people affected, and test other weights**

Two points apart is too close to call: a small change in the weights could swap them.